

Week 3

Find your own examples: _____

2×2 matrices, A , D , and C such that $A \neq 0$, $C \neq D$, but $AC = AD$. _____

Here are some examples:

$$\begin{matrix} & A & & C & & AC \\ \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} & \begin{pmatrix} 1 \\ 1 \end{pmatrix} & = & \begin{pmatrix} 0 \\ 0 \end{pmatrix} \end{matrix}$$

$$\begin{matrix} & A & & D & & AD \\ \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} & \begin{pmatrix} 2 \\ 2 \end{pmatrix} & = & \begin{pmatrix} 0 \\ 0 \end{pmatrix} \end{matrix}$$

or

$$\begin{matrix} & A & & C & & AC \\ \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} & \begin{pmatrix} 1 & 1 \end{pmatrix} & = & \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \end{matrix}$$

$$\begin{matrix} & A & & D & & AD \\ \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} & \begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix} & = & \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \end{matrix}$$

Question: can you find examples where $AC \neq 0$?

Explain why if $AB = AC$ and A^{-1} exists, then $B = C$.

If $AB = AC$, multiply both sides from left by A^{-1} , then

$$A^{-1}AB = A^{-1}AC$$

$$\Rightarrow IB = IC$$

$$\Rightarrow B = C.$$

Give an example of matrices, A, B such that neither A nor B equals zero and yet $AB = 0$.

Examples in the first problem work.

In general, we want to find a nontrivial solution to $AX = 0$. Think about

for what A such a solution exist. Pick such an A and find a solution.

Note that X can have any number of columns.

Using the inverse of the matrix, find the solution to the systems

$$\begin{aligned} \textcircled{1} \begin{pmatrix} 1 & 0 & 3 \\ 2 & 3 & 4 \\ 1 & 0 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} &= \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \quad \textcircled{2} \begin{pmatrix} 1 & 0 & 3 \\ 2 & 3 & 4 \\ 1 & 0 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} \\ \textcircled{3} \begin{pmatrix} 1 & 0 & 3 \\ 2 & 3 & 4 \\ 1 & 0 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} &= \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \quad \textcircled{4} \begin{pmatrix} 1 & 0 & 3 \\ 2 & 3 & 4 \\ 1 & 0 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 3 \\ -1 \\ -2 \end{pmatrix}. \end{aligned}$$

$$A = \begin{pmatrix} 1 & 0 & 3 \\ 2 & 3 & 4 \\ 1 & 0 & 2 \end{pmatrix} \rightsquigarrow A^{-1} = \begin{pmatrix} -2 & 0 & 3 \\ 0 & 1/3 & -2/3 \\ 1 & 0 & -1 \end{pmatrix}.$$

If $AX=b$, then $X=A^{-1}b$. Hence

$$\textcircled{1} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -2 & 0 & 3 \\ 0 & 1/3 & -2/3 \\ 1 & 0 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 7 \\ -4/3 \\ -2 \end{pmatrix}$$

$$\textcircled{2} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -2 & 0 & 3 \\ 0 & 1/3 & -2/3 \\ 1 & 0 & -1 \end{pmatrix} \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -4 \\ 1/3 \\ 2 \end{pmatrix}$$

$$\textcircled{3} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -2 & 0 & 3 \\ 0 & 1/3 & -2/3 \\ 1 & 0 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ -2/3 \\ 0 \end{pmatrix}$$

$$\textcircled{4} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -2 & 0 & 3 \\ 0 & 1/3 & -2/3 \\ 1 & 0 & -1 \end{pmatrix} \begin{pmatrix} 3 \\ -1 \\ -2 \end{pmatrix} = \begin{pmatrix} -12 \\ 1 \\ 5 \end{pmatrix}$$

Now give the solution in terms of a, b , and c to

$$\begin{pmatrix} 1 & 0 & 3 \\ 2 & 3 & 4 \\ 1 & 0 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}.$$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -2 & 0 & 3 \\ 0 & 1/3 & -2/3 \\ 1 & 0 & -1 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} -2a+3c \\ \frac{1}{3}b-\frac{2}{3}c \\ a-c \end{pmatrix}$$

Show that if A is an invertible $n \times n$ matrix, then so is A^T and $(A^T)^{-1} = (A^{-1})^T$.

We need to show $(A^{-1})^T$ acts like the inverse of A^T . Then by the uniqueness of the inverse of a matrix, it is the inverse of A^T . That is

we need to check

$$\begin{cases} \textcircled{1} A^T \cdot (A^{-1})^T = I \\ \text{and} \\ \textcircled{2} (A^{-1})^T \cdot A^T = I \end{cases}$$

$$\textcircled{1} A^T \cdot (A^{-1})^T = (A^{-1} \cdot A)^T = I^T = I \quad \checkmark$$

because $B^T C^T = (CB)^T$

$$\textcircled{2} (A^{-1})^T A^T = (AA^{-1})^T = I^T = I \quad \checkmark$$

Thus, $(A^{-1})^T$ is the inverse of A^T , that is,

$$(A^T)^{-1} = (A^{-1})^T$$

Show $(AB)^{-1} = B^{-1}A^{-1}$ by verifying that $\textcircled{1} AB(B^{-1}A^{-1}) = I$ and $\textcircled{2} B^{-1}A^{-1}(AB) = I$.

(Hint: if A^{-1} exists for an $n \times n$ matrix, then it is unique. That is, if $BA = I$ and $AB = I$, then $B = A^{-1}$.)

$$\textcircled{1} (AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AIA^{-1} = AA^{-1} = I \quad \checkmark$$

$$\textcircled{2} (B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B = B^{-1}IB = B^{-1}B = I \quad \checkmark$$

Since $(B^{-1}A^{-1})$ acts like the inverse of AB , by uniqueness of the inverse, it is the inverse of AB , that is

$$(AB)^{-1} = B^{-1}A^{-1}.$$

Show that $(ABC)^{-1} = C^{-1}B^{-1}A^{-1}$ by verifying that $(ABC)(C^{-1}B^{-1}A^{-1}) = I$ and $(C^{-1}B^{-1}A^{-1})(ABC) = I$.

Similar to the previous problem:

$$\begin{aligned}(ABC)(C^{-1}B^{-1}A^{-1}) &= AB(\overbrace{CC^{-1}}^{=I})B^{-1}A^{-1} \\ &= A(\overbrace{BB^{-1}}^{=I})A^{-1} \\ &= AA^{-1} \\ &= I\end{aligned}$$

$$\begin{aligned}\text{and } (C^{-1}B^{-1}A^{-1})(ABC) &= C^{-1}B^{-1}(\overbrace{A^{-1}A}^{=I})BC \\ &= C^{-1}(\overbrace{B^{-1}B}^{=I})C \\ &= C^{-1}C \\ &= I\end{aligned}$$